

Time Transfer by Laser Link: Data analysis and validation to the ps level

Pierre Exertier^{a,*}, E. Samain^a, N. Martin^a, C. Courde^a, M. Laas-Bourez^a, C. Foussard^a,
Ph. Guillemot^b

^aUMR Geoazur, Université de Nice, Observatoire de la Côte d'Azur, 250 rue A. Einstein, F-06560 Valbonne, France

^bCNES, French Space Agency, 18 avenue E. Belin, F-31455 Toulouse, France

Received 10 July 2014; received in revised form 13 August 2014; accepted 14 August 2014

Available online 27 August 2014

Abstract

The Time Transfer by Laser Link (T2L2) is a very high resolution time transfer technique based on the recording of arrival times of laser pulses at the satellite. T2L2 was designed to achieve time stability in the range of 1 ps over 1000 s and an accuracy better than 100 ps. The project is in operation onboard the Jason-2 satellite since June 2008. The principle is based on the Satellite Laser Ranging (SLR) technology; it uses the input of 20–25 SLR stations of the international laser network which participate in the tracking. This paper focuses on the data reduction process which was developed specifically to transform the raw information given by both space instrument and ground network: first to identify the triplets (ground and onboard epochs and time of flight of the laser pulse), second to estimate a usable product in terms of ground-to-space time transfer (including instrumental corrections), and thirdly to produce synchronization between any pair of remote ground clocks. In describing the validation of time synchronizations, the paper opens a way for monitoring the time difference between ultra-stable clocks thanks to a laser link at a few ps level for Common View passes. It highlights however that without accurately characterizing the onboard oscillator of Jason-2 and knowing the unavailability of time calibrations of SLR stations generally, time transfer over intercontinental distances remain difficult to be accurately estimated.

© 2014 COSPAR. Published by Elsevier Ltd. All rights reserved.

Keywords: Time transfer; Laser ranging

1. Introduction

The performance of atomic clocks have advanced of several order of magnitude these last two decades. Microwave frequency standards exhibit accuracy in the 10^{-16} range (Parker, 2010) optical clocks already demonstrated stability of a few parts in 10^{-18} . In order to build a universal time scale the real performance limitation are not the clocks but rather the ability to accurately transfer the time and frequency information from one place to another. Clock comparison techniques are currently based on microwave

systems; they use the Global Navigation Satellite System (GNSS), and in particular the Global Positioning System (GPS) Common View (CV) (Ray and Senior, 2005) and the Two-Way Satellite Time and Frequency Transfer (TWSTFT) (Jiang and Petit, 2009). Actually, the international UTC/TAI time and frequency transfer network is currently based on these two independent space techniques.

The development of techniques for the comparison of distant clocks and for the distribution of stable and accurate time scales will have significant impact for future space and ground international activities (navigation, broadband communication networks and fundamental science). For this reason, the comparison of techniques is very important to make progress in terms of efficiency, precision and

* Corresponding author.

E-mail address: pierre.exertier@geoazur.unice.fr (P. Exertier).

finally accuracy (Piester et al., 2008 and Delporte et al., 2008). In recent years, we are witnessing the development of optical time and frequency transfer technique over distances up to several hundred km in fibers (Predehl et al., 2012). In this context, the development of free space laser is necessary and even logical on intercontinental distances.

The Time Transfer by Laser Link project (T2L2) is a satellite time transfer technique based on the Satellite Laser Ranging (SLR) technology (Samain et al., 2008). It was launched on the Jason-2 satellite (at 1,335 km altitude) in June 2008, and it has run continuously since that time. If the concept existed since the 90s – i.e., the LASSO experiment (Fridelance and Veillet, 1995) – T2L2 is currently the only time transfer technique based on a free laser with expected performance of very high quality: a time stability of 1 ps at 1000 s and 10 ps at 1 day, and an accuracy better than 100 ps (Weick et al., 3123). For several decades, the SLR technique contributed very significantly to geodetic and geodynamic research (Exertier et al., 2006). As one of the fundamental space geodetic techniques, SLR provides measurements of distances between ground laser stations and satellites with both high precision (tens of ps) and accuracy (Degnan et al., 1993). Nowadays, many satellites are equipped with laser retro-reflectors including Jason-2. Routine laser observations in the global network are coordinated by the International Laser Ranging Service (ILRS) which has been fully operational since 1998 (Pearlman et al., 2002). The ILRS also combines products from different analysis centers; it makes a unique contribution building fundamental terrestrial references, such as the Earth geocenter and scale factor of the reference frame (Altamimi et al., 2011).

T2L2, which therefore uses the international ILRS network, is an opportunity to evaluate the laser ranging system and to develop the necessary ground operations (hardware and software) to achieve a time transfer of metrological level between ultra-stable ground remote clocks (Samain et al., 2011). In the current context, the laser ranging facilities do not have the high performance (accuracy) of atomic clocks. However, among the 20–25 participating SLR stations, 8 are equipped with an ultra-stable atomic clock (hydrogen-maser or cesium clock) as a local reference timescale synchronized with UTC.

The purpose of this paper is to describe the data analysis conducted as part of an Analysis Center (AC) developed specifically to compute the time transfer by laser from all the data sources. The paper addresses the methods, the data quality (in terms of time transfer), the system efficiency (laser ranging), and the products (time transfer and frequency bias). Special attention is given to the error budget, in particular to define the accuracy provided by the T2L2 technique.

In Section 2, we briefly describe the ground and space segments of the mission, that is useful in understanding the capability of T2L2 (advantages and disadvantages). Then in Section 3, we present the basic methodology to

perform the ground-to-space time transfer including principle, study of noise, and products. From these products we describe in Section 4 the ground-to-ground time transfer technique (in CV) in its principle and validation and we discuss the short- and long-term stability and accuracy. The Section 5 is dedicated to the ground-to-ground time transfer in non CV, by beginning with the estimate of the frequency bias of the onboard oscillator and its modeling over some thousands of seconds. We conclude with Section 6.

2. Space and ground segments

The Jason-2 satellite is equipped with a Laser Reflector Array (LRA) of 10 cm diameter, in support of the precise orbitography, in addition to the Doppler Orbitography and Radio-positioning Integrated by Satellite (DORIS) tracking system (Cerri et al., 2010; Willis et al., 2006). Due to the altitude of the mission, of 1335 km, the satellite is flying for around 1000 s over a given station on the ground (at an elevation of around 20 degrees). The passes are successive (every two hours) with a maximum of 6–7 passes that can be observed daily by a station. As a passenger of the Jason-2 mission, T2L2 uses the LRA to determine the 2-way link and the DORIS Ultra Stable Oscillator (USO) as the onboard frequency reference.

2.1. The space instrument

The complete instrument model is described in Samain et al. (submitted for publication). We briefly describe below the three main components that enter into the data reduction process (from raw data to corrected time transfer) and its associated error budget: (i) the detection module, (ii) the geometry between the LRA and T2L2 optics, and (iii) the reference clock associated to the event timer.

Detection. T2L2 is able to detect a part of the received energy (at 532.1 nm) of very short laser pulses (width is tens of ps) that are emitted by any ground SLR station. The detection module is used to grab the laser pulse and to transmit the signal to the timer. T2L2 has a linear detector which allows us to make a measure of the received energy and to discern laser pulses from earth optical background. The detection chain includes a nonlinear photodiode for chronometry; it works in the single- and multi-photons regimes. This photodetector is based on an avalanche photodiode operating in Geiger mode; it has a time walk dependency on the received energy and, to a lower part, on the attitude (Vrancken et al., 2008).

Geometry. The collimating optics (linear and nonlinear) and the LRA are not located at the same place, the reflection and detection points do not coincide. It is therefore necessary to compute, for each received laser pulse, the distance (of around 110 mm) between these points. The field of view of the optics is 55 degrees. Given the altitude of the mission, this allows the system to see, at the same time, ground remote station 5000 km (see Fig. 1).

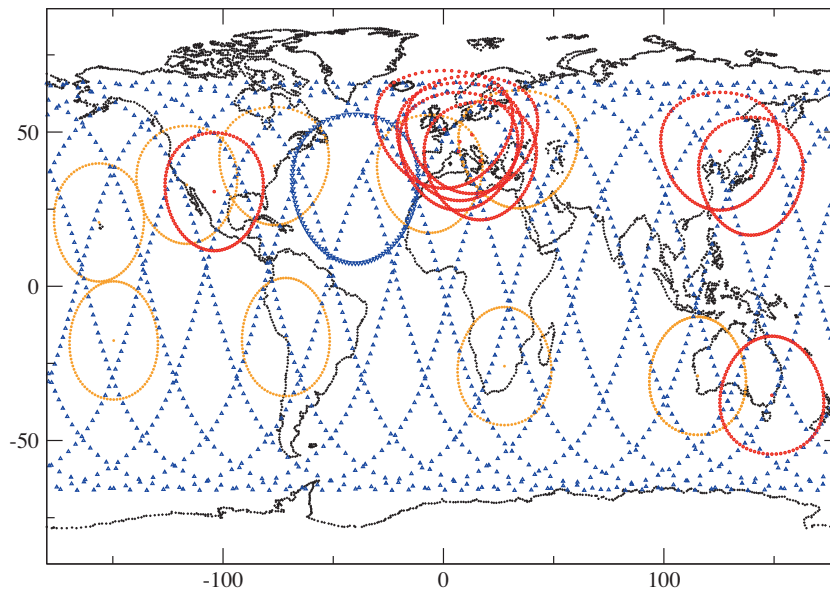


Fig. 1. Location of SLR stations involved in the T2L2/Jason-2 tracking. Visibility circles at 20 degrees; circle in red indicates sub-nanosecond performance. The blue circle is an example of the field of view of T2L2.

Clock. The received signal is dated thanks to the T2L2 event timer which has a resolution of 1 ps and a repeatability error of 3 ps rms (Samain et al., submitted for publication). The onboard frequency reference uses the DORIS oscillator; it is a quartz crystal whose best stability is of a few parts in 10^{-13} at 1000 s (Auriol and Tourain, 2010). As a consequence of the long term stability of this USO, the best performance of the T2L2 time transfer is expected rather in the Common View (CV) mode, both in terms of time stability and accuracy. In return, T2L2 thus offers the opportunity to “read” (locally) and even monitor (in the long term) the frequency bias of DORIS.

Acquisition. The T2L2 space instrument is working permanently; it has an electronic dead time of 200 μ s that corresponds to an acquisition rate of 5000 Hz as a maximum. The Jason-2 satellite being co-visibility of several SLR stations over some areas (like Europe and Asia), T2L2 is able to acquire a great number of the laser events per unit time. In addition, optical events due to sun and earth albedo enter in the T2L2 detection chain. This adds noise to the measurements (up to a few ns) and means to select data carefully. All the data are downloaded from T2L2/Jason-2 to the ground with a classical microwave link; the files are distributed on a daily basis by the Centre National d’Etudes Spatiales (CNES), in Toulouse. Fig. 2 shows the number of optical events per day which have been acquired by T2L2 since June 2008. Some peaks (300,000 data points) are due to SLR stations that operate a kHz laser.

2.2. The ground network

Stations. The ground segment of T2L2 is based on the SLR network. Around 35–37 SLR stations currently

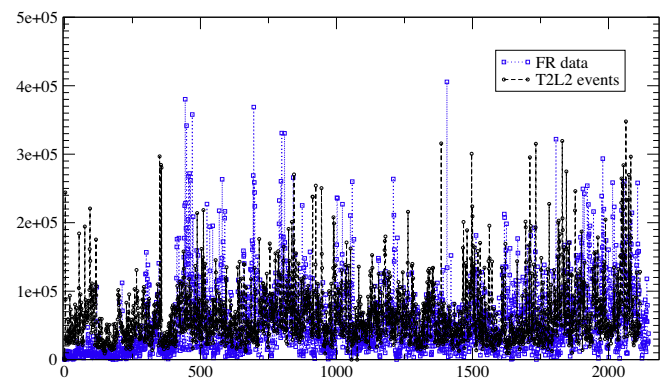


Fig. 2. Number of both optical T2L2 events and available laser ranging data per day since the Jason-2 launch (June 2008). The number of onboard events can be greater than the ground laser data (FR: Full Rate), the number of participating stations to the T2L2 project being smaller than the number of stations firing Jason-2 as a support for precise orbitography.

participate in the tracking of the Jason-2 satellite (Cerri et al., 2010). Among these, 20–25 stations provide us with SLR data regularly. Fig. 1 shows the geographical coverage relative to a one-day orbital arc of Jason-2. Moreover, Fig. 3 shows the number of stations per day that are participating in the T2L2 project since June 2008.

SLR data. SLR stations provide us with Full Rate (FR) data regularly via some international data centers such as the European Data Center (EDC) in Munich, Germany and the Crustal Dynamics Data Information System (CDDIS) at NASA/Goddard. Fig. 2 shows the number of FR data (in blue) which have been downloaded per day from EDC and/or CDDIS since the beginning of the project. The delay in getting the data varies from 1 h or less to – on occasion – several days.

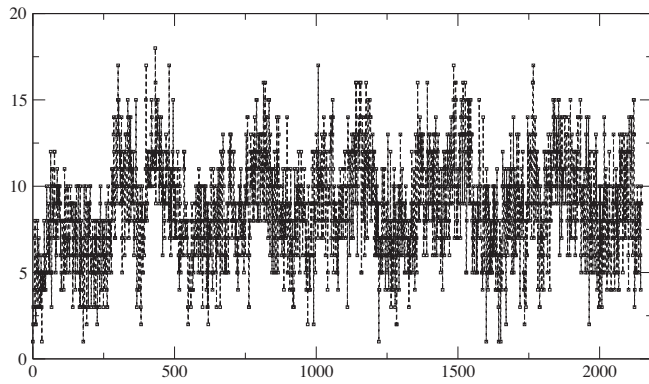


Fig. 3. Number of SLR stations involved in the T2L2 tracking per day since the Jason-2 launch (June, 2008).

FR data consist of both the emission date (resolution is 1 ps, but sometimes 100 ns, depending of the station) and the 2-way time of flight (see the CRD format on the ILRS website). All stations which are identified by T2L2 use a 532 nm laser with a repetition rate of the laser of 10 Hz (sometimes kHz).

Table 1 shows that the number of passes per station is roughly constant along the time. For SLR stations that contribute significantly to the T2L2/Jason-2 tracking, this number is around 400 over a 6-month period (summer and autumn). The average number of onboard epochs

per pass (nBd-P) that were recorded by T2L2 per pass is very homogeneous, *i.e.* for MOBLAS systems (there is a marked increase just after 2008). We should note that weather, atmosphere transmission, and tuning of each ground laser are sometimes factors that limit the data acquisition onboard. As a consequence, laser pulses with a low energy level ($<0.1 \mu\text{J m}^{-2}$, measured by T2L2) are difficult to detect and many materialize only from time to time (see the ratio “% T” on Table 1).

Clocks. The emission date of laser pulses is given by the reference time system of the station. In the long term, the local timescales are synchronized by GPS to avoid time offsets greater than 100 ns relative to UTC. Laser ranging facilities do not have the high performance of atomic clocks generally. Actually, the time stability and accuracy of the reference clocks used at ground level are very different from one station to another. But it is clear that what matters above all in the satellite tracking are the short-term stability (<1 s) and frequency accuracy. For space geodesy, we should note that an accuracy of measurement epochs of few hundred ns relative to UTC is enough to ensure satellite along-track position and/or absolute positioning to a precision of a few mm (Sośnica et al., 2012).

Since the beginning of the T2L2 project, 8 SLR stations are equipped with an ultra-stable clock such as an hydrogen maser (H-maser) or a cesium notably in Asia, Europe, and the United States (see Table 2). The frequency stability

Table 1

Statistics of data for two 6-month periods (July to December), in 2008 and 2013; Eng: average onboard energy ($\mu\text{J m}^{-2}$), nP: number of passes, nPval: number of validated passes (>50 triplets), nBd-P: average number of onboard data points per pass, % T: average ratio of the number of triplets *vs.* the number of ground echoes.

ILRS n., station	Eng	2008				2013			
		nP	nPval	nBd-P	% T	nP	nPval	nBd-P	% T
1824 Kiev						47	0	0	0
1873 Simeiz	0.57					105	0	0	0
1893 Katzively	0.68	48	20	206	48	113	5	208	32
7080 Mac Donald	1.94	31	16	218	82	33	26	275	65
7090 Yarragadee	5.55	449	424	562	94	434	389	829	92
7105 Greenbelt	3.86	185	163	544	87	142	132	1367	77
7110 Monument Peak	5.53	61	56	616	82	135	133	1484	83
7119 Mt Haleakala	11.56					78	23	114	69
7124 Papeete	6.09	18	18	602	89	47	45	649	79
7237 Changchun ^(*)	0.13	239	54	1079	24	428	171	23335	3
7308 Koganei Tokyo	0.71					27	13	730	23
7403 Arequipa	9.78	64	38	146	97	124	87	190	89
7501 Hartebeesthoek	1.62	40	29	419	87	132	123	1479	57
7810 Zimmerwald	0.68	288	263	1713	49	478	440	3322	27
7821 Shanghai						2	0	0	0
7824 San Fernando	1.03	94	58	605	33	73	42	293	50
7825 Mt Stromlo STL3	0.23	284	57	678	13	250	4	3135	6
7832 Riyadh		75	58	602	56				
7838 Simosato	1.56					61	50	344	49
7840 Herstmonceux	0.42	229	148	1223	41	224	173	1543	16
7841 Potsdam	0.51	44	25	747	23				
7845 Grasse MeO	2.07	19	14	1024	48	430	422	1076	59
7829 FTLRS Grasse	0.36	5	5	502	37	2	2	180	27
7941 Matera MLRO	1.53	78	77	660	91	200	193	963	70
8834 Wettzell	1.25	133	111	394	80	205	167	647	44

^(*) kHz laser; FTLRS: French Transportable Laser Ranging Station.

achieved by these clocks is of 10^{-15} at 1000 s. Finally, some stations, notably the block of MOBLAS systems, provide us with epochs to 100 ns.

3. From ground to space

Schematically, the ground to space time transfer is deduced from the difference between the arrival time of the laser pulse (in the timescale of T2L2/Jason-2) and the start time (in the station timescale) shifted by the time of flight divided by two. The first characteristics of the ground to space time transfer such as noise, precision and time stability were described in Exertier et al. (2010).

The situation described in Section 2 implies that the data processing procedures must be very robust (in the mathematical sense) to accommodate all the cases (laser stations, clocks, technological changes, data quality and quantity, etc.). We provide below all the improvements that were made thanks to newly developed algorithms.

3.1. Principle

We consider below a set of three data (leave out a “triplet”): (i) the measured 2-way time of flight tof , (ii) the emission date t_e of each laser pulse (the return date being t_r), and (iii) the arrival date t_b on board Jason-2. The time transfer equation between ground and space clocks is as follows, for a station s and at time (t) over one pass, without any correction:

$$tof = (t_r - t_e)$$

$$D(t_e) = \frac{1}{2}(tof - C_{\text{Sagnac}})$$

$$\Delta_s^b(t) = t_b - [t_e + D(t_e)] \quad (1)$$

where:

$D(t)$ is computed from tof , a quantity which is calibrated from the geodetic reference point of the station: the cross-axes of its telescope,

$C_{\text{Sagnac}} = \frac{2}{c^2} \mathbf{D}(t_e) \cdot \dot{\mathbf{X}}(t_e)$: is the effect due to Earth rotation, which is computed at t_e from the coordinates $\mathbf{x}$ and $\mathbf{X}$ of the satellite and station in a geocentric reference system (inertial), respectively; $\mathbf{D}(t_e) = \mathbf{x} - \mathbf{X}$, $\Delta_s^b(t)$ represents the desynchronization between both clocks along the pass (t_0 will be used as the date of the first triplet).

The triplets are obtained under the following conditions: (i) the 3 basic data are all available, (ii) tof does not exist but t_b does; if t_e is available in the data file of the considered station, then $D(t_e)$ can be estimated thanks to a local interpolation. This permits us to increase the number of triplets by a factor of 1.2 and to decrease the noise on tof (Exertier et al., 2010; Fridelance and Veillet, 1995). The basis of the T2L2 data analysis consists in identifying the triplets (tof, t_e, t_b) for each pass of each station. This is done by making a correlation between onboard and ground epochs shifted by $|\mathbf{D}(t_e)|$ at the $1 \mu\text{s}$ level. The procedure also allows us to identify “true” triplets and to reject outliers (false epochs coming from solar events or other sources abnormally detected by T2L2). At this stage obviously, we do not consider any correction, and for practical reasons $D(t)$ is replaced by $|\mathbf{D}(t)|$ which quantity comes from the precise orbit of Jason-2 (Cerri et al., 2010).

Table 1 shows the average ratio of the number of triplets of ground echoes (% T). The quantity is roughly correlated with the average energy of laser pulse which are received on board. Table 2 shows the relative quality of raw triplets (without instrumental corrections) by station; after a linear fit of Eq. (1), the rms of residuals per pass were averaged over two 6-month periods, in 2008 and 2013. First, SLR stations which use an ultra-stable clock are clearly at the sub-ns level. Secondly, technological changes (reference clock) made at least in Wettzell (in 2009), Zimmerwald (in 2009), and Changchun (end of 2008) have a great and direct impact on the quality of time transfer. Thirdly, SLR systems that provide measurement epochs at 100 ns cannot be competitive (*i.e.*, to precisely estimate the onboard frequency bias); inversely, T2L2 is thus an opportunity to synchronize their clocks from other SLRs that use an ultra-stable clock.

After applying the instrumental corrections to raw triplets, the time transfer equation evolves as:

Table 2
Ground to space time transfer (in ps); mean values of rms per pass (after a linear fit) over two 6-month periods, in 2008 and in 2013. First block: SLR stations equipped with an ultra-stable clock (except 7825); Second block: other cases, notably with measurement epochs at 100 ns.

Station, ILRS number	Clock	2008	2013
7080 Mac Donald	Cesium	4111	580
7237 Changchun	H-maser	3419	1223
7308 Koganei	Atom. fountain	114	759
7810 Zimmerwald	H-maser, 2009-	24,403	177
7825 Mount Stromlo		2990	2969
7840 Hersmonceaux	H-maser	1208	100
7841 Potsdam		20,227	
	(in 2011)		2400
7845 Grasse MeO	Cesium, –2010	133	
	H-maser, 2010-		123
7829 FTLRS Grasse	id.	216	134
7941 Matera	H-maser	170	197
8834 Wettzell	H-maser, 2009-	22,985	122
	Resolution 100 ns		
7090 Yarragadee	59,251		
7105 Greenbelt	57,591		
7110 Monument Peak	62,354		
7119 Mt Haleakala	41,010		
7124 Papeete	38,124		
7403 Arequipa	57,904		
7501 Hartebeesthoek	57,904		
7824 San Fernando	41,441		
7838 Simosato	28,958		

$$\Delta_s^b(t) = [t_b + C_{T \text{ walk}}] - (t_e + C_{\text{Cal}, s}) - D(t_e) - C_{T2L2-LRA} + \int C_{F \text{ shift}} dt + \Delta_s^b(t_0) \quad (2)$$

where:

$C_{T \text{ walk}}$: is the instrumental correction due to time walk of the nonlinear photodiode; it is a time shift in the avalanche process induced by the photon number variation. Its amplitude ranges from 30 to 150 ps and depends on the received energy. This kind of photodetector has been using for several years by the laser ranging community (i.e., Degnan et al. (1993) and Nicolas et al. (2000)). We will consider below the corrected date: $\tilde{t}_b = t_b + C_{T \text{ walk}}$ thanks to the numerical model (time walk compensation *vs.* energy and attitude) given by Samain et al. (submitted for publication),

$C_{T2L2-LRA}$: is the geometric correction between the LRA and T2L2 optics, which strongly depends on the satellite attitude in relation to instantaneous vector $\mathbf{D}(t)$. Its typical amplitude is of roughly 0.2 ns. It is computed from the coordinates of both LRA and T2L2 optics, quantities that are provided by the Jason-2 project at the < 1 mm level in the satellite frame (Cerri et al., 2010). On the other hand, the attitude of Jason-2 is measured by stellar sensors; it is issued in the form of a set of quaternions available every minute in the J2000 reference system. It provides attitude angles at a precision of ± 0.1 degrees. We will consider in the following, the corrected date: $\tilde{t}_e = t_e + D(t_e) + C_{T2L2-LRA}$,

$C_{F \text{ shift}} = 1/c^2(W + 1/2 \dot{x}^2)$: is the relativistic frequency shift following the Schwarzschild solution, where W is the gravitational potential at $\mathbf{x}$, and $\dot{x}$ the satellite velocity. The periodic part of this effect is in the orbital period and its submultiples. The integrated effect causes a signal (linear and nonlinear) of 40–100 ps over one pass.

$C_{\text{Cal}, s}$: is the time calibration of the laser station s : that is, the total delay (in cables, electronic devices, etc.) between the reference point of the time and frequency laboratory (pps unit) and the geodetic reference point used for the range (cross-axes of the telescope). Actually, this quantity has to be measured by some hardware procedure (Samain et al., 2013).

The identification of the triplets allows us to determine the local geometry and the angle of incidence of the laser beam on the T2L2 detector. These informations are mandatory to compute the instrumental corrections above mentioned; the time evolution of Eq. (2) is thus analyzed according to:

$$\Delta_s^b(t) = \tilde{t}_b - \tilde{t}_e + [\Delta_s^b(t_0) - C_{\text{Cal}, s}] + \int C_{F \text{ shift}} dt \quad (3)$$

For practical reasons, we simplified this equation in two parts. The first one is linear, what is of unlike frequency between clocks on board and on the ground. The second one corresponds to the residuals (named $\delta_s(t)$). Now,

Fig. 4 is a typical example of a ground-to-space time transfer, where instrumental corrections were applied at this stage. It shows the residuals after a linear fit was applied.

3.2. Noise and signals

The study of instrumental noise had been made before the launch of the T2L2 project, from theoretical and empirical approaches (Vrancken et al., 2008; Weick et al., 3123).

Now, by taking into account all the data sources (including correction models) to be used, a detailed study of the time stability of the time transfer equation was conducted. This study was needed to shed light on certain issues and answer some questions such as: the role and efficiency of the laser technique, some aspects about time and frequency metrology on board and at ground level, etc. This led us to separate noise and signals in the time transfer equation – Eq. (3) – under the following form:

$$\delta_s(t) = N(t) + S_{\text{nl}}(t) + E_\delta \quad (4)$$

where: $N(t)$ are white noise, $S_{\text{nl}}(t)$ are remaining nonlinear signals, and E_δ are constant errors. For practical reasons, the term $\delta_s(t)$ was inserted as a numerical quantity that represents the noise and errors of $\Delta_s^b(t)$ of Eq. (3). Let us briefly describe the different sources which disrupt the calculation of the ground-to-space time transfer:

$C_{T \text{ walk}}$: the correction model has a precision of around 3–35 ps (the worst case occurs at low energy flow) (Vrancken et al., 2008). As a result of the design of T2L2 space instrument, the used metrology for the detection is similar, in its principle, to that used by ground stations (Degnan et al., 1993). We considered that respective noise are thus comparable being a few tens of ps ($N_{\text{Detections}}$).

$C_{T2L2-LRA}$ varies due to the change of geometry [station *vs.* satellite] during the flight of Jason-2 above the station. The overall limit of this geometric correction is estimated to be 2–3 ps. Obviously, the residual signal (S_{Geometry}) is at low frequency.

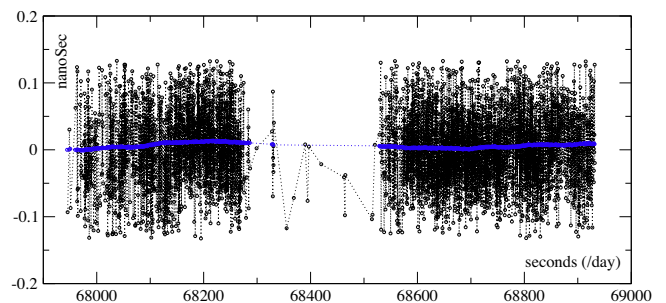


Fig. 4. Ground to space time transfer from Grasse SLR station to T2L2/Jason-2, over one pass. Black dots are the residuals (in ns) of Eq. (3) after removing a linear fit. Blue dots are the corresponding residuals after applying a low-pass filter at 75 s. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Dates: t_e and t_b are measured at the 5–10 ps and 3 ps levels (N_{Dates}) with respect to the ground and onboard event timers, respectively. The time stability of the series of ground epochs t_e (S_{Dates}) is estimated at a few ps over 1000 s when a H-maser associated to a high-resolution event timer are used. On the other hand, stability of the series t_b (S_{USO}) strongly depends on the USO frequency stability (Auriol and Tourain, 2010).

C_{Sagnac} : the correction is calculated from both the Jason-2 ephemerides and the ITRF 2008 solution of station coordinates (Cerri et al., 2010; Altamimi et al., 2011). As a result of their respective precision, of 3–5 cm for the orbit and 1–2 cm for station coordinates, the uncertainty on the calculation of C_{Sagnac} is of 1–2 ps; the remaining signal S_{Sagnac} is at low frequency.

$C_{\text{F shift}}$: the relativistic signal (frequency shift) is integrated over the satellite pass. We consider the time variations of both the gravitational potential and the satellite velocity; on the other hand we do not consider the rotation of the Earth (station velocity). The remaining signal ($S_{\text{Relativity}}$) is of 1–2 ps at low frequency.

Considering the differences of technologies between SLR stations in addition to the T2L2 performance, the overall noise ($N(t)$) of the ground-to-space time transfer is at the level of 25–60 ps (std deviation) (Exertier et al., 2010). The remaining nonlinear effects ($S_{\text{nl}}(t)$) which were estimated thanks to this study are at the level of several tens of ps over one pass. The constant errors (E_δ) which were estimated in an empirical (numerically) way, are in the range of 5–10 ps.

$$N(t) = [N_{\text{Detections}}^2 + N_{\text{Dates}}^2 + \dots]^{1/2}$$

$$S_{\text{nl}}(t) = S_{\text{Geometry}} + S_{\text{Dates}} + S_{\text{USO}} + S_{\text{Sagnac}} + S_{\text{Relativity}}$$

Fig. 5 is a typical example of the time deviation (TDEV) plot for a ground-to-space time transfer of around 600–700 s duration. The SLR system (Grasse) is using an H-maser which is the reference clock of the station. It is

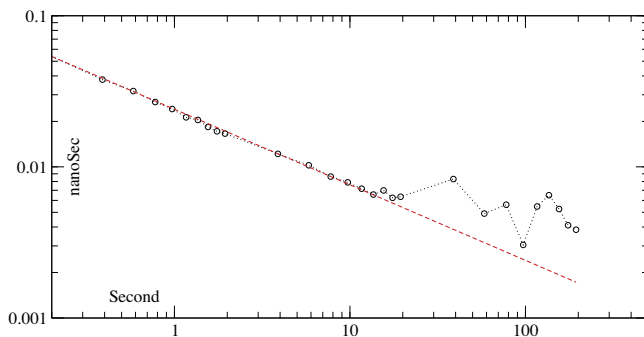


Fig. 5. Allan time deviation (in ns) of the time transfer residuals as shown in Fig. 4 (after applying a linear fit). Black dots come from the triplets, whereas red dots come from a theoretical value decreasing in $1/\sqrt{t}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

found that at $\tau = 0.1$ s the starting Allan deviation is 50–60 ps, which confirms the estimated std deviation of the points individually.

After 1 s, the TDEV signal is decreasing as $\tau^{-1/2}$ up to 6–8 ps at around 75 s. Many different situations were analyzed; Fig. 5 shows the minimum of Allan deviation which was obtained per pass over 1 year as a function of τ for Grasse (F), Herstmonceux (UK), and Yarragadee (AUS). It follows that, statistically, we can say that the best time stability of ground to space time transfer is a few ps at $\tau = 75$ s, provided the ground time system of the transmitting laser station is ultra-stable.

Beyond 100 s, the effect introduced first by the USO and, to a lesser extent, by the other remaining nonlinear effects produce a growing uncertainty of several tens of ps. This low frequency noise (such as random walk noise) can be quite large numerically over a larger and larger group of data (i.e. in case of non Common View time transfer).

On the other hand when a station does not use a high resolution event timer (1 ps) and/or an ultra-stable clock, the ground-to-space time transfer performance decreases to a standard deviation of 2–3 ns (for Mt Stromlo, Potsdam, Changchun) up to 57–62 ns for the worst cases (see Table 2). The slope of the Allan deviation is (also) in $\tau^{-1/2}$ and the minimum is at 0.2 to 5 ns at $\tau = 120$ s, respectively. Fig. 6 shows the smallest TDEV which was obtained per pass over 1 year as a function of τ for Yarragadee. On the basis of this statistics, we adopted an average time stability of 2–3 ns at $\tau = 120$ s for that type of situation.

Due to a factor of 1000 between the best case (at 50–60 ps) and the worst case (55 ns), we developed a dedicated robust analysis of the time transfer equation. A key point being to consider *a priori* parameters, per station, that help the analysis to automatically adapt to different situations. The great interest is to make a clean rejection

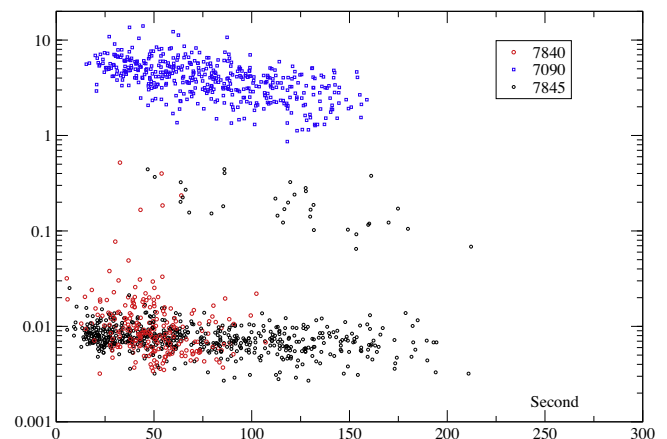


Fig. 6. Statistics (over 1 yr) of the lowest Allan time deviations (ns) of ground to space time transfers calculated for Yarragadee (7090, blue), Herstmonceux (7840, red), and Grasse (7845, black) SLR stations as a function of τ (s). Yarragadee station has a resolution of 100 ns of its date. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of outliers (around 20%), whatever cases we have, in order to stabilize the time transfer equation and to save data (triplets).

3.3. Basic products

The output of the ground-to-space time transfer estimation process is something important in order to give access to computing ground-to-ground time transfer. It is constrained by two main factors: the temporal resolution, and the necessity to mix datasets (triplets per station). Actually the idea was to develop, once and for all, a series of synchronization Δ_s^b per station. The goal was to enable efficient retrieval of common time transfer in the series, at any time and for any pair of ground stations.

We adopted a 1-s resolution which is the result of multiple factors: the average number of available triplets per pass, the sequence of epochs $t_b(s)$ (the number of points per second varies from 8–9 to 1–2 and until 0.02 in case of hole), and the fact that there are two stations in set to be combined. In the same logic, we developed a filtering procedure in order to eliminate white noise in Eq. (3):

$$\langle \Delta_s^b(t_j) \rangle_F = \frac{1}{\sum_i W_{ij}[w_i]} \sum_i W_{ij}[\Delta_s^b(t_i) w_i]$$

$$W_{ij} = (1 - A)^\tau \quad \text{with } \tau = \frac{|t_i - t_j|}{F} \quad \text{and } A = 0.6 - 0.8$$

where:

$\Delta_s^b(t_j)$: is the raw triplet value, and j (or i) run from 1 to the total number (n_T) of available triplets per pass,

W_{ij} : is a weighting function which depends on A (constant) and F (the width of the filter),

w_i : is a weight per data point $[0, 1]$; it is worth 1, *a priori*, but evolves in 0 following the selection criterion ($n \times \text{RMS}$, $n = 2-3$) of triplet values,

In order to define F and A in W_{ij} , we analyzed many data sets (triplets) with a view to reduce the amount of noise and thus characterizing the small signals $S_{nl}(t)$ relative to the noise (factor of 10). We chose the average minimum of the Allan deviation plots, that is $F = 75$ s for high-performance time transfers (ps level) and $F = 120$ s for the other cases (ns level, see Table 2). We chose also $A = 0.7$.

For practical reasons, the series were calculated to the round second in the onboard timescale. The method consists in interpolating the quantity $\langle \Delta_s^b(t) \rangle_F$ over 1 to 2 s in order to calculate a synthetic $\langle \Delta_s^z \rangle$ which corresponds to a round second t_b^z . The corresponding $t_e^z(s)$ is then computed by accounting for an approximate linear variation of $\langle \Delta_s^b(t) \rangle_F$ over the small interval (of 1–2 s). We will see below that the resulting error which affects $t_e^z(s)$ do not exceed 1–2 ps, because the hypothesis of linearity is verified (with a relative error of 10^{-13} to some 10^{-12}) on this small duration. When the number of triplets per second is >2 , the procedure account for the date $t_b(s)$ closest to t_b^z . As a result of this process, Table 3 shows both outputs (as part of the

series) for the Herstmonceux and Grasse SLR stations (in January 2013). The series include the t_b^z epochs (T2L2 count) and the corresponding $\tilde{t}_e^z(s)$ (MJD and Second (day)) in the timescale of station s .

4. From ground to ground (CV)

The method to compute the ground-to-space time transfer is robust and forms the basis for all subsequent steps. The computation of the ground-to-ground time transfer between 2 stations $s1$ and $s2$ is based on the independent estimation of both time transfers Δ_s^b to the satellite. During a CV time transfer, the effects of the onboard clock (USO) are removed over times greater than the interval separating the measures. We will see below that other common effects can be removed in order to accurately estimate the ultimate performance of the time transfer data processing (see validation of analysis).

4.1. Principle

The principle consists to remove the effect of the USO by establishing the difference $\Delta_{s1}^b(t) - \Delta_{s2}^b(t)$; actually, we look for a direct estimate of the difference of epochs $-\tilde{t}_e(s1) + \tilde{t}_e(s2)$.

To this end, it is necessary to determine an exact synchronization between the two series of corresponding epochs $t_b(s)$, as they are not simultaneous in the on board timescale. As a result of calculating 1-s series of epochs $\tilde{t}_e^z(s)$ for each ground-to-space time transfer, the ground-to-ground desynchronization is directly obtained by differencing of $\tilde{t}_e^z(s)$ at all common $\tilde{t}_b^z$ seconds along the pass. The main advantage of this method is to avoid making other calculations and/or interpolations and assumption about the USO behavior. The total desynchronization between both ground clocks can be expressed as:

$$\Delta_{s2}^{s1}(t) = -\tilde{t}_e^z(s1) + \tilde{t}_e^z(s2) + \Delta_{s2}^{s1}(t_0) + [-C_{\text{Cal}, s1} + C_{\text{Cal}, s2}] \quad (5)$$

where:

$-\tilde{t}_e^z(s1) + \tilde{t}_e^z(s2)$: is the time difference between ground clocks. This quantity slowly evolves over time essentially due to the difference of frequency (Ψ_s) between both oscillators, the difference of noise and local time delays in cables, etc.

$\Delta_{s2}^{s1}(t_0)$: is the desynchronization at the beginning of the common pass (date t_0 of the first common triplet). It contains some phase offsets notably which can be applied by any station (irregularly),

$-C_{\text{Cal}, s1} + C_{\text{Cal}, s2}$: is the difference in the time calibrations of $s1$ and $s2$.

Fig. 7 shows an example of CV time transfer between the Grasse and Herstmonceux SLR stations; both were equipped with an H-maser clock since 2010. The

Table 3

Typical products of the ground-to-space time transfer processing: example of the 1-s series for both Herstmonceux (7840, pass 1) and Grasse (7845, pass 3) SLR stations January, 3 2013. nb: is the number of triplets per second, and Error: is the formal error of date $\tilde{t}_e(s)$ in ps.

MJD	Second (day)	T2L2 count (s)	nb	Error (ps)	Hersm.
56303	45546.461660823158	49135447	2	1.9	
56303	45548.461660731030	49135449	1	3.4	
56303	45553.461660500708	49135454	1	3.4	
56303	45559.461660224323	49135460	1	3.4	
56303	45560.461660178259	49135461	2	2.2	
					Grasse
56303	45547.461660416468	49135448	2	0.9	
56303	45548.461660370404	49135449	5	0.6	
56303	45549.461660324339	49135450	5	0.6	
56303	45552.461660186147	49135453	8	0.4	
56303	45553.461660140083	49135454	9	0.3	
56303	45554.461660094019	49135455	6	0.3	
56303	45558.461659909763	49135459	7	0.4	
56303	45559.461659863699	49135460	9	0.5	
56303	45560.461659817634	49135461	9	0.6	

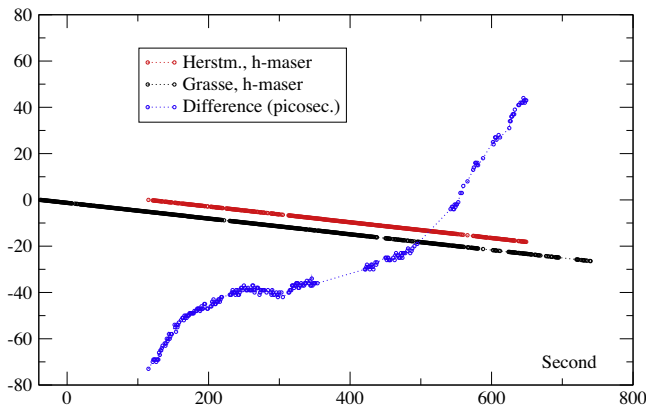


Fig. 7. Time transfer in Common View between the Grasse and Herstmonceux stations (on June, 25 2010); the zero is arbitrary. In black and red, each ground-to-space time transfer in μs . In blue, the ground-to-ground time transfer in ps. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ground-to-space passes (black and red, in μs) are very close, being affected by the same local relative frequency bias of the USO: $v_b/v_0 \simeq 33.6 \times 10^{-9}$ on June, 25, 2010 – where v_0 is 10 MHz for DORIS. The time difference (blue dots in ps; the zero is arbitrary) shows a signal which can be described as:

- a linear trend of around 60 ps over 600 s ($\approx 10^{-13}$) which is mainly due to the difference of frequency ($\Psi_{s1} - \Psi_{s2}$) between both ground clocks,
- nonlinear effects which can be summarized by the sum ($[S_{nl}]_{s1} + [S_{nl}]_{s2}$), where:

$$[S_{nl}]_s = S_{\text{Geometry}} + S_{\text{Dates}} + S_{\text{Relativity}} \quad (6)$$

where are taken into account for each station the errors due to the geometry, to the laser link (date, range, and ground delays) and, to the lesser level, to the relativistic correction (effect of Earth's rotation on the station).

4.2. Validation of analysis

The validation of the different methods of analysis which were specifically developed for the Time Transfer by Laser Link project is of a major importance considering the expected performance of T2L2. The purpose of this section is to report the validation in terms of time stability and accuracy.

Several campaigns were conducted at the Grasse Geodetic fundamental Observatory between 2010 and 2013. Here, we focused on two campaigns conducted in 2010 and then in 2013. The particularity of these two campaigns was to involve the two SLR systems that are available at the Observatory, both of them being synchronized by the same H-Maser.

In addition to the Grasse SLR station (MeO, 7845) we deployed the mobile laser ranging station (FTLRS, 7829) in the same building (30 meters long). The FTLRS has a telescope of 13 cm diameter and its 10 Hz laser has an energy per pulse of around 25–35 mJ (Nicolas et al., 2000). As a result of the telescope size and laser energy, MeO was able to provide much more available triplets per second compared to FTLRS (see Tables 3 and 4).

Both campaigns were conducted independently but in the same way in order to be compared. During these campaigns, taking into account the importance to highlight the ultimate performance of T2L2 (data and methods of analysis), ground cables were installed to connect FTLRS and MeO to the H-maser of Grasse.

Over one pass. Due to the choice of the instrument configuration, several sources of noise and signals were removed. First, we used the same clock at ground; cables (to connect both stations) were at the same temperature (in the same building). Second, the geometry of both passes was the same, as all the relativistic corrections (satellite and station positions and velocities). These two campaigns permitted us to establish pure comparisons between both

Table 4

Characteristics of both stations and data, during the CV time transfer campaigns at the Grasse Geodetic Observatory (4 days from May, 22 2010, and 14 days from January, 2 2013).

	Comment	FTLRS	MeO	FTLRS	MeO
Telescope	Diameter (m)	0.13	1.54	0.13	1.54
Laser	Energy (J)	0.030	0.150	0.030	0.150
	Cadence (Hz)	10.	10.	10.	10.
Clock		H-maser	H-maser	H-maser	H-maser
		2010		2013	
Data	nb of passes	13	53	11	37
	nb of shots	92,284	287,903	100,134	336,043
	nb of triplets	8314	50,457	2,546	131,158
	nb of CV passes	10		11	
	nb of epochs (1 s)	990		538	
	rms of CV	9 ps		11 ps	

lasers via T2L2 the clock signals being removed as at ground level as in space (CV).

Fig. 8 shows one of the CV passes which was observed on May, 23 2010 (in ns; zero is arbitrary). Compared to the previous case (CV between Grasse and Herstmonceux, see Fig. 7), respective signals by station and difference have the same type of behavior. Obviously for the Grasse CV there is no drift in the difference. Thus, it is a good example to show common low frequency noise observed by both stations (in black and green) on the one hand, and the remaining nonlinear signals (S_{nl}) in the difference (in blue) on the other. Here, it is essentially due to both lasers (difference in pulse width and detector), and to the slightly different response of the nonlinear diode of T2L2 to the two types of pulses (width and energy).

Fig. 9 (bottom plot) shows all the CV passes (in ns) which are plotted according to seconds during the pass. This plot is interesting from the standpoint of short-term time stability. On average, signals are almost the same; the average rms of CV passes is of 9 ps. At this level, the origin of the nonlinear behavior was studied from hardware experiments essentially (Samain et al., 2013).

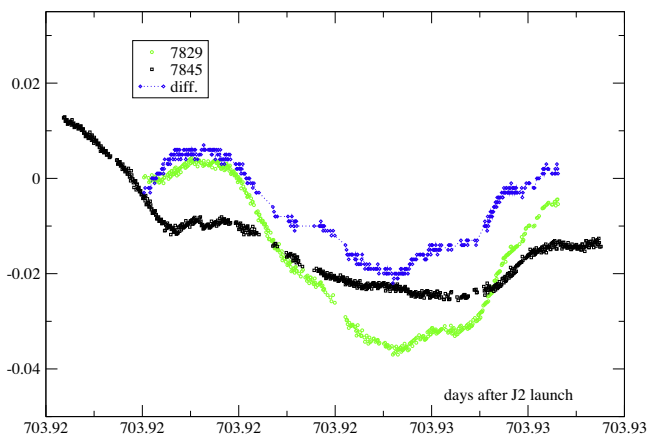


Fig. 8. Ground-to-ground time transfer (ns) between the MeO (black) and FTLRS (green) Grasse SLR stations; both SLR systems used the same reference clock (H-maser). Difference is in blue, and zero is arbitrary (on May, 23 2010). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

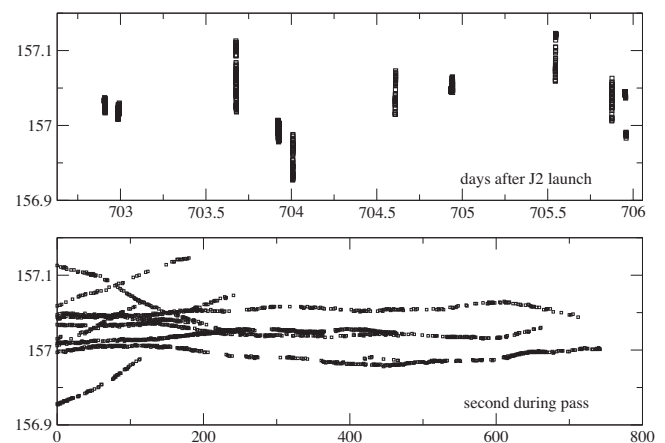


Fig. 9. Ground-to-ground CV time transfers (in ns) over 4 days (May 2010) between MeO and FTLRS in Grasse; both SLR systems use the same reference clock (H-maser). Top: successive passes along time (days). Bottom: CV passes are plotted according to second.

Difference by pass. As a consequence of very short CV passes and large interruptions, yielding a series of clock data with many gaps, a suitable analysis was necessary to be developed in order to define Normal Points (NP) per pass and their associated error budget. On a long-term basis obviously the 1-s resolution provided by each CV time transfer has to be reduced in order to provide an accurate series of T2L2-NP (for a given link) to be compared to other techniques like *i.e.* the Global Positioning System (GPS) used for time transfer in CV (Ray and Senior, 2005).

The 2010 campaign lasted four days (22–25th May); our objective was to characterize the ability of the T2L2 data analysis to calculate the ground-to-ground link (the time difference between both laser stations) in the same conditions over several CV passes. Due to the unhomogeneity of the data, the number n_T of available CV triplets range from 5 to 400 per pass, the duration T_{CV} of each CV pass is not the same and, finally, the local error (rms) varies slightly (see Tables 5 and 6). For these reasons, we established an empirical law ($[2T_{CV} \times n_T] / [\text{rms} \cdot 10^5]$) in order to estimate the weight of each individual CV pass. Fig. 9 (top plot) and Table 5 give the main results of the 2010 CV campaign. From the T2L2-NP, the weighted average

Table 5

Results of the 2010 CV time transfer campaign. NP: are Normal Points (weighted values of the CV ground-to-ground time transfer) computed per pass. The rms of NP is 0.025 ns, and the average is 157.030 ns.

Stations & pass	MJD (sec in day)	NP (ns)	rms (ns)	nb	Dur. (s)	Weight
7829 2–7845 4	55338 78500.0	157.027	0.007	220	466.0	0.220
7829 3–7845 5	55338 85463.0	157.020	0.006	428	661.0	0.428
7829 2–7845 2	55339 58497.0	157.057	0.030	130	284.0	0.064
7829 3–7845 4	55339 79862.0	156.994	0.009	397	742.0	0.397
7829 1–7845 1	55340 717.0	156.951	0.019	62	113.0	0.049
7829 2–7845 2	55340 52582.0	157.045	0.017	56	241.0	0.051
7829 4–7845 5	55340 81176.0	157.052	0.005	303	712.0	0.303
7829 2–7845 2	55341 47151.0	157.093	0.022	43	180.0	0.030
7829 4–7845 6	55341 75669.0	157.035	0.016	41	113.0	0.039
7829 5–7845 7	55341 82617.0	157.037	0.016	109	487.0	0.102

Table 6

Results of the 2013 CV time transfer campaign. The rms of NP is 0.039 ns, and the average is 188.609 ns.

Stations & pass	MJD (sec in day)	NP (ns)	rms (ns)	nb	Dur. (s)	Weight
7829 1–7845 1	56294 68465.0	188.641	0.009	57	554.0	0.057
7829 2–7845 2	56294 75419.0	188.638	0.011	127	574.0	0.127
7829 1–7845 1	56299 39940.0	188.642	0.022	5	30.0	0.003
7829 1–7845 2	56301 42627.0	188.671	0.005	41	235.0	0.041
7829 2–7845 3	56302 72337.0	188.578	0.005	26	253.0	0.026
7829 1–7845 2	56303 38145.0	188.600	0.029	15	209.0	0.008
7829 2–7845 6	56303 67007.0	188.581	0.004	51	309.0	0.051
7829 1–7845 1	56304 32751.0	188.555	0.018	18	610.0	0.015
7829 2–7845 2	56304 39636.0	188.521	0.007	14	97.0	0.014
7829 1–7845 1	56306 35498.0	188.593	0.006	83	614.0	0.083
7829 1–7845 1	56307 65256.0	188.648	0.055	101	672.0	0.028

value of the ground-to-ground link is 157.030 ± 0.025 ns (rms). It corresponds to the mean time delay between both stations over four days. Finally, the repeatability of this experiment is estimated thanks to the absolute mean deviation; for the present series we found 0.042 ns, thus establishing a time stability (TDEV) of 10 ps over one day.

Accuracy. From the 2013 campaign (it was conducted from December 2012 to January 2013, actually) we extracted 14 days (January 2–15, 2013) in order to have available 11 CV passes that could be compared to the 2010 statistics (see Table 4). Our objective was to estimate the accuracy that is provided by the present analysis. To this end, the time calibration of both SLR systems was conducted in January 2013 thanks to dedicated metrology resulting notably in a ground “distance” of $188.546 \text{ ns} \pm 0.035$ between both stations (Samain et al., 2013).

From the T2L2 data analysis, we calculated NP and then the weighted average value of the ground-to-ground link (see Table 6). The result is 188.609 ± 0.039 ns (the repeatability is estimated at 0.040 ns over 14 days). The difference between the hardware and software procedures is thus of 0.063 ns (absolute value).

Let us summarize the part of constant errors (E_δ) that are addressed by the present data processing. By rejecting the errors that were removed thanks to the configuration of the CV campaign, the total error is, as a maximum, the sum (absolute sense) of multiple factors. Several solutions were calculated by taking into account the errors due to the geometry (2–3 ps), to the relativity (1–2 ps),

and to the different processing steps (such as interpolation, filtering, etc.). We found, as a maximum, an error of 15 ps. In addition, the constant part of the errors, of technological source (such as: timer, detector, range, etc.), have to be accounted for in order to establish a proper error budget. To this end, a dedicated study will be conducted by involving dedicated metrology (i.e. Samain et al. (2013)).

5. The non Common View case

Due to the altitude of the T2L2/Jason-2 project and the present geographical coverage of the SLR network, CV time transfers are limited to some areas like Asia, Europe and the United-States. If the best performance of T2L2 can be achieved for CV passes (see previous section), there is great interest in demonstrating the capability of the laser technique to transfer time over inter-continental distances.

Obviously, the best frequency stability of the DORIS oscillator being at $3 \cdot 10^{-13}$ over 1000 s (Auriol and Tourain, 2010), the expected performance of the ground-to-ground time transfer over “distances” of 5000–10,000 s (one orbital revolution) should be at a level of tens of ns. In addition, temperature change (even <1 degree in space) results in frequency shifts and phase drifts becoming outdated over several revolutions.

Actually, we do not have available a very fine model of the USO frequency which could be integrated to achieve a ground-to-ground time transfer at the ns level over one revolution of Jason-2. It is thus important to find ways to

establish such a model at least locally. In this section, we give a first example of nonCV time transfer based on an empirical (numerical) model of the USO frequency variations. We describe below the method and give an error budget through some examples.

5.1. Estimate of the USO relative frequency

In the previous Section 3, we detailed the error budget of a ground-to-space time transfer essentially when the ground reference time system is based on an ultra-stable clock. The local time stability of Eq. (3) ($\Delta_s^b(t)$) allowed us to precisely estimate a time drift over a short duration (around 1 min), according to this approximation of the derivative:

$$\frac{v_s^b(t)}{v_0} = \frac{1}{F} \left[\Delta_s^b\left(t + \frac{F}{2}\right) - \Delta_s^b\left(t - \frac{F}{2}\right) \right] \quad (7)$$

It corresponds to a relative frequency offset v_s^b/v_0 seen by the station s during the time interval F with respect to the nominal frequency v_0 of the USO. Its precision was expected at a few parts in 10^{-13} ($d v_s^b/v_0$) provided the number of triplets (nbT) associated to the duration of the pass were sufficiently large enough (nbT > 50, and pass duration > 100 s). We chose to deliver a series of values for $F = 60$ s which are coherent at best and completely unexamined at worst. As a by product of the ground-to-space time transfer process, we established a data base with relative frequency offsets v_s^b/v_0 by station s .

As explained by several authors, each DORIS receiver has its proper time evolution of the frequency. Thanks to the precise orbitography, it is possible to monitor the long-term drifts of the different DORIS onboard oscillators with respect to their nominal frequency (Lemoine and Capdeville, 2006). As a result of the regular SLR tracking of T2L2/Jason-2 since July 2008, the data base allows us to monitor the DORIS relative frequency variations. For example, the USO mean (sense approximate) relative frequency was $+20 \cdot 10^{-9}$, in July 2008 with a drift of around $+0.025 \cdot 10^{-9}$ per day decreasing to $+0.008 \cdot 10^{-9} \text{ d}^{-1}$ in December 2013, as a result of the USO time history (age).

The series of frequency offsets provided by the T2L2 data was compared to the DIODE/Jason-2 estimations (Jayles et al., 2010). The agreement is of $1.2 \cdot 10^{-12}$ in term of relative frequency (Ch. Jayles, private communication, DORIS group for mission and performance, CNES, December 2013). The 2013 campaign served as an opportunity to calculate the relative frequency offset v_s^b/v_0 thanks to both SLR stations (MeO and FTLRS) connected the same clock at ground. The comparison was based on 11 CV passes (see Table 6). It gave an average difference of $1\text{--}2 \cdot 10^{-13}$.

5.2. Modeling of the onboard timescale

The “real” behavior of an oscillator in space is very complex. As an example of this, see the modelization of

the quartz crystal that was developed to maintain v/v_0 at the level of 10^{-12} for the DORIS/Jason-1 USO (Lemoine and Capdeville, 2006). Here, the Jason-2 USO is better in term of frequency stability (Auriol and Tourain, 2010). In addition to random walk, its behavior is certainly affected by several physical processes which are mainly at the orbital period (as temperature, radiations, etc.).

Because the objective of the present study is to achieve the calculation of nonCV time transfers at a few ns over 5000 to 10,000 s, we established a mean numerical solution rather than a physical model. For these reasons, we adopted the following form:

$$v^b(t)/v_0 = [C_v \cos \omega t + S_v \sin \omega t] + a_v + b_v t + c_v t^2 + d_v t^3 + \dots + C_{F \text{ shift, satellite}} + C_{F \text{ shift, station}} \quad (8)$$

where $(C, S; a, b, c, d, \dots)_v$ are coefficients to be estimated by a least-squares procedure, ω is the orbital pulsation, and $C_{F \text{ shift}}$ are the relativistic corrections calculated from the position and velocity of the satellite and from the altitude of each station, respectively.

The present modeling involved several SLR stations (see Table 7), one of them serving as a reference (weight = 1) was the Grasse station with an average of 3 passes a day in January 2013. After adjusting the model at order 8 over 31 days, the overall std deviation of the residuals was $0.53 \cdot 10^{-12}$ in term of relative frequency; 1997 data (on 2531) were selected after a 2σ selection procedure. This result was not too dependent on the considered period; the data were selected properly and the model applied over 15 days, 20 or 25 days. The weight for each station was estimated empirically.

The estimated biases per station (mean of model residuals) were of a reasonable level; they did not exceed a few parts in 10^{-13} in many cases. We should note that, being extracted from the ground-to-space equation the frequency offsets v_s^b/v_0 were influenced also by the frequency Ψ_s of the ultra-stable oscillator used at ground level.

The Fig. 10 shows the residuals after the model was fitted; the zero and the global trend are arbitrary. Now, the question that had to be addressed was: is it possible to use this model, to be integrated over time, to achieve a

Table 7

DORIS relative frequency offsets and std dev. (rms) estimated by SLR stations (in 10^{-12}), relative to a numerical model fitted over 31 days (from January, 1 2013). nb-P: is the number of passes; the number (nb) of v_s^b/v_0 depends on the elimination criteria (at 2σ); wgt: are the weights per station.

Station, ILRS number	nb-P	nb v_s^b/v_0	wgt	bias	w-rms
7845 Grasse	75	1584/ 1452	1.0	0.08	0.51
7829	19	228/ 213	1.0	0.04	0.51
7840 Herstmonceux	11	49/ 32	0.5	−0.24	0.48
7308 Koganei	7	34/ 16	0.7	0.50	0.59
7080 Mac Donald	6	44/ 15	0.4	−0.58	0.69
7941 Matera	19	278/163	0.5	0.03	0.57
8834 Wettzell	8	70/ 40	0.5	−0.04	0.54
7090 Yarragadee	42	278/ 52	0.1	−0.08	4.47/ 0.44

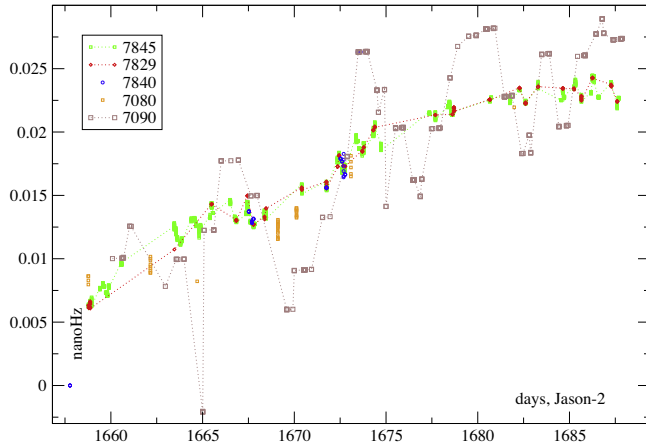


Fig. 10. Relative frequency offsets estimated by T2L2, for several passes of several SLR stations in January 2013. Residuals after removing approximative bias and drift. SLR's are: Grasse (7845 and 7829), Herstmonceux (7840), Mac Donald (7080), and Yarragadee (7090).

ground-to-ground time transfer over intercontinental distances (that is around 5000 s).

5.3. Time transfer in nonCV

The integration of $v^b(t)$, between two epochs t_0 to t used a 1-min step starting from a given phase $\tilde{t}_e(s1, t_0)$. Relativistic effects on frequency ($C_{F \text{ shift}}$) due to the varying gravitational potential and velocity of the Jason-2 satellite were included. The nonCV time transfer can be represented by:

$$\Delta_{s2}^{s1} = [-C_{\text{Cal}, s1} + C_{\text{Cal}, s2}] + \tilde{t}_e(s1, t_0) + \int_{t_0}^t v^b(t) dt + \int_{t_0}^t C_{F \text{ shift}} dt - \tilde{t}_e(s2, t) \quad (9)$$

where t_0 is a given date $\tilde{t}_e$ of station $s1$ to be integrated over time, the result being compared to the closest date $\tilde{t}_e$ of station $s2$. The current process was established such as:

- to choose available ground-to-space passes between two SLR stations,
- to avoid calculating the nonCV time transfer for a period larger than 6:00,
- to start the integration of the model at the first date of the chosen pass of station $s1$,
- to stop the integration at the end of the chosen pass of station $s2$.

On the other hand, the present study showed that the model should be constrained in order to avoid divergence of the numerical integration over several thousands of seconds. The constraints were estimated thanks to successive ground-to-space time transfer passes observed at a given place $s1$ at 2:00 or 4:00. Actually, the integration process of $v^b(t)$ between two successive passes of Jason-2 at station $s1$ should give the same time offset as the one which is obtained by making the difference of ground time over

the same period (we assume no consequent phase drift of the ground clock over few hours). If not, a frequency constraint is calculated to slightly modify the initial model.

Several tests were made (see Table 8). In each case, the Grasse SLR 7845 was used as the Master clock $s1$ in order to determine the constraints at 2:00 or 4:00. As a consequence, the passes of the chosen stations $s2$ were necessarily situated between two Grasse passes. The epochs $\tilde{t}_e$ of Grasse were time calibrated by the quantity $C_{\text{Cal}, s1}$. Due to a permanent time transfer by a calibrated GPS link to the “Observatoire de Paris”, the H-maser of Grasse is regularly shifted by some frequency offsets to the UTC timescale at around 5 ns or less (Laas-Bourez et al., 2013). As a result of this, the present nonCV time transfer from Grasse to another SLR station $s2$ allowed us to determine the time offset of clock $s2$ relative to UTC plus the time calibration $C_{\text{Cal}, s2}$. The present study were not dedicated to the accuracy of the nonCV time transfer (because $C_{\text{Cal}, s2}$ was unknown), but was rather used to estimate the efficiency of the method (model, constraints, and processing).

For the nonCV time transfer to Yarragadee, several passes were calculated thanks to the well-known tracking capability of the station. Due to the drift of the local clock, it was not possible to directly compare the results from one pass to another, at the ns level. Using a 2:00 or 4:00 integration for the same pair of passes gave a measure of the precision, whereas comparing successive passes of $s2$ from $s1$ gave an estimate of the repeatability. Both were at the 2–3 ns level.

For the nonCV time transfer to Mac Donald and Koganei, we investigated our data base for December, 2012. As a result of the initial model adjustment over 25 days, the v_s^b were filtered by regression yielding to a global precision of 0.57×10^{-12} , which confirms the homogeneity of both data and approach. Now, the results (Table 8) are better than for Yarragadee because both SLR stations are equipped

Table 8

Non Common View time transfers from Grasse (7845) to Yarragadee (AUS), to Mac Donald (US), and to Koganei (J) in December 2012 and January 2013. Int.: is the duration of the integration; Synchro. (in ns): is the offset between Grasse and the considered SLR station, and Err.: is the formal error.

Name, ILRS ID	MJD & Time	Int. (h)	Synchro. & Err. (ns)
Yarragadee, 7090 (100 ns)	56299 13:40	2	−1188.30 ± 2.0
	15:54	4	−1191.10 ± 1.5
	56309 11:43	2	−1131.52 ± 6.4
		4	−1132.19 ± 6.2
	13:40	4	−1130.56 ± 1.2
	56314 11:43	2	−1120.79 ± 0.4
Mac Donald, 7080 (Cesium)		4	−1121.23 ± 0.4
	56269 22:40	2	15.73 ± 0.2
		4	15.87 ± 0.1
	56300 17:01	4	27.41 ± 0.4
Koganei, 7308 (Atomic fountain)		6	26.88 ± 0.4
	56265 8:05	17	−268.22 ± 0.6
	56271 6:41	13	−270.38 ± 0.1
	8:22	13	−273.23 ± 0.5

with an ultra-stable clock. Error sources can be roughly separated as: the error emanating from the integration process, which is estimated at around 0.3–0.5 ns over 5000 s, whereas the time data (ground to space products $\tilde{t}_e$) are at 0.1–0.3 ns.

6. Conclusion

The T2L2 Analysis Centre was developed to understand the data (space instrument, SLR data, time and frequency metrology between ground and space), to operate the data treatment on a daily basis, to help the project in characterizing the performance, and to promote some scientific applications. Important work was necessary to be completed in terms of analysis in order to achieve an operational data processing. T2L2 is the first space time transfer technique based on a laser, with metrological performances and data sources of different quality (and quantity) coming from the current ILRS network. The article presents the methods of data analysis, validations and error budgets, in both Common View (CV) and non Common View cases.

As a consequence of very short CV passes and large interruptions, yielding a series of clock data with many gaps, a suitable analysis was necessary to be developed in order to extract as much information as possible from the T2L2 data. This analysis is based primarily on the calculation of the ground-to-space time transfer. The time stability is established at a few ps at 75 s for SLR stations that are equipped with an ultra-stable clock (H-maser, generally). Thanks to dedicated data sets, the validation of analysis allowed us to confirm the consistency between the data processing and the expected accuracy of T2L2 (<100 ps, and probably around 50–70 ps).

Considering T2L2 as a space time transfer technique which runs permanently, the long-term time stability provided by the technique is thus a key factor. The important point which is demonstrated here, is the fact that T2L2 data analysis is able to produce ground-to-ground NP (in CV) with the same very high quality, at any time.

In order to enhance the T2L2 project and the effort of ILRS stations (around 20–25) to provide FR data, we developed a dedicated website (<https://t2l2.oca.eu/>). Before all, the website was designed to provide time transfer data to the community. A part of it concerns the data available (ground and onboard), the identified “triplets” and their performance by satellite pass and by station. It can be used for many different purposes and, above all, the estimation of ground-to-space and ground-to-ground (CV) time transfers between any pair of available stations.

The SLR technique, thanks to T2L2, provide a time coverage (monitoring) of the Jason-2 USO frequency variations which is much better in quality than in quantity relative to the DORIS capability. It is due to the current geographical coverage of SLR stations and to the availability of ultra-stable clocks at ground level. This later is the main limitation of the project actually for nonCV time

transfer. Nevertheless, some time calibration campaigns could be very useful to the community in providing more occasions to permanently synchronize all SLR stations to the same timescale at the same period.

Acknowledgments

We thank the Centre National d’Etudes Spatiales for providing constant help in the development of the Analysis Center and for conducting campaigns. We thank the laser community which helped us in providing FR data on Jason-2, since 2008. Finally, special thank is addressed to the Grasse SLR team for observing Jason-2 regularly and participating in many campaigns and studies.

References

- Altamimi, Z., Collilieux, X., Métivier, L., 2011. ITRF2008, an improved solution of the international terrestrial reference frame. *J. Geod.* 85 (8), 457–473.
- Auriol, A., Tourain, C., 2010. DORIS system: the new age. *Adv. Space Res.* 46, 1484–1496.
- Cerri, L., Berthias, J.P., Bertiger, W.I., Haines, B.J., Lemoine, F.G., Mercier, F., Ries, J.C., Willis, P., Zelensky, N.P., Ziebart, M., 2010. Precision orbit determination standards for the Jason series of altimeter missions. *Mar. Geod.* 33 (S1), 379–418.
- Degnan, J.J., 1993. Millimeter accuracy satellites for two color ranging. In: *Proceedings of the Eighth International Workshop on Laser Ranging Instrumentation*, (SEE N94-15552 03-19), 16 p.
- Delporte, J., Mercier, F., Laurichesse, D., Galy, O., 2008. GPS carrier-phase time transfer using single-difference integer ambiguity resolution. *Int. J. Navig. Obs.* 2008. Art. ID 273785.
- Exertier, P., Bonnefond, P., Deleflie, F., Barlier, F., Kasser, M., Biancale, R., Ménard, Y., 2006. Contribution of laser ranging to Earths sciences. *C.R. Geosci.* 338 (14–15), 958–967.
- Exertier, P., Samain, E., Bonnefond, P., Guillemot, Ph., 2010. Status of the T2L2/Jason2 experiment. *Adv. Space Res.* 46 (12), 1559–1565, DORIS Special Issue, Precise Orbit Determination and Applications to Earth Sciences, P. Willis (Ed.).
- Fridelance, P., Veillet, Ch., 1995. Operation and data analysis in the LASSO experiment. *Metrologia* 32 (1), 27–33.
- Jayles, Ch., Chauveau, J.P., Roza, F., 2010. DORIS/Jason-2: better than 10 cm on-board orbit available for near real-time altimetry. *Adv. Space Res.* 46, 1497–1512.
- Jiang, Z., Petit, G., 2009. Combination of TWSTFT and GNSS for accurate UTC time transfer. *Metrologia* 46 (3), 305–314.
- Laas-Bourez, M., Samain, E., Courde, C., Oneto, J.L., Exertier, P., Rovera, D., Abgrall, M., Fridelance, P., Guillemot, Ph., 2013. Time and Frequency distribution improvement in Calern/Geoazur laboratory for T2L2 campaigns. In: *Proceedings of the EFTF & International Frequency Control Symposium (EFTF/IFC) 2013*, July 21–25, Prague, Republik Tchèque, pp. 869–872.
- Lemoine, J.-M., Capdeville, H., 2006. A corrective model for Jason-1 DORIS Doppler data in relation to the South Atlantic Anomaly. *J. Geod.* 80 (8–11), 507–523.
- Nicolas, J., Pierron, F., Kasser, M., et al., 2000. French transportable laser ranging station: scientific objectives, technical features, and performance. *Appl. Opt.* 39, 402–410.
- Parker, T.E., 2010. Long-term comparison of caesium fountain primary frequency standards. *Metrologia* 47 (1).
- Pearlman, M.R., Degnan, J.J., Bosworth, J.M., 2002. The international laser ranging service. *Adv. Space Res.* 30 (2), 135–143.
- Piester, D., Bauch, A., Breakiron, L., Matsakis, D., Bianzano, B., Koudelka, O., 2008. Time transfer with nanosecond accuracy for the realization of international atomic time. *Metrologia* 45 (2), 185.

- Predehl, K., Grosche, G., Raupach, S.M.F., Droste, S., Terra, O., Alnis, J., Legero, Th., Hänsch, T.W., Udem, Th., Holzwarth, R., Schnatz, H., 2012. A 920-Kilometer optical fiber link for frequency metrology at the 19th decimal place. *Science* 336 (6080), 441–444.
- Ray, J., Senior, K., 2005. Geodetic techniques for time and frequency comparisons using GPS phase and code measurements. *Metrologia* 42 (4), 215–232.
- Samain, E., Weick, J., Vrancken, P., 2008. Time transfer by laser link – the T2L2 experiment on JASON-2 and further experiments. *Int. J. Mod. Phys. D* 17 (7), 1043–1054.
- Samain, E., Exertier, P., Guillemot, Ph. et al., 2011. Time transfer by laser link – T2L2: current status and future experiments. In: *Proceeding of the European Frequency and Time Forum*, May 1–5 2011, San Francisco, California, USA, pp. 378–383.
- Samain, E., Courde, C., Exertier, P. et al., 2013. Time transfer by laser link – T2L2: microwave link comparison. In: *Proceeding of the 18th International Workshop on Laser Ranging*, November 11–15 2013, Fujiyoshida, Japan, 13-Po47.
- Samain, E., Vrancken, P., Guillemot, Ph., Fridelance, P., Exertier, P., accepted for publication. Time transfer by laser link (T2L2): characterization and calibration of the flight instrument, *Metrologia*, MET-100164.R1.
- Sošnica, K., Thaller, D., Jaggi, A., Dach, R., Beutler, G., 2012. Sensitivity of Lageos orbits to global gravity field models. *Artif. Satellites* 47 (2), 47–65.
- Vrancken, P., 2008. Characterization of T2L2 (time transfer by laser link) on the Jason 2 ocean altimetry satellite and micrometric laser ranging, Thèses. Université Nice Sophia Antipolis (23/09/2008), Etienne Samain (Dir.), [oai:tel.archives-ouvertes.fr:tel-00367703] – <<http://tel.archives-ouvertes.fr/tel-00367703>>.
- Weick, J., Samain, E., Vrancken, P., Guillemot, Ph., 2007. Prospective performance budget of the T2L2 experiment. In: *Proc. IEEE Frequency Control Symposium, Joint with the 21st European Frequency and Time Forum*, May 29–June 1, pp. 312–314.
- Willis, P., Jayles, C., Bar-Sever, Y., 2006. DORIS, from altimetric missions orbit determination to geodesy. *C.R. Geosci.* 338 (14–15), 968–979.